



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024

CS2101-1- COMPUTER ORGANIZATION AND ARCHITECTURE
(CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which is the proper order for the computer functions cycle?
 A) Processing, input, output, storage B) Input, processing, storage, output
 C) Input, output, processing, storage D) Input, processing, output, storage
2. The media and methods used to keep computer information available for later use are called...
 A) Computer processing B) Computer output
 C) Computer input D) Computer storage
3. Which is equivalent to 1 GB
 A) 1,024 Terabytes B) 1,024 Kilobytes
 C) 1,024 Megabytes D) 1,024 Bytes
4. Which flag indicates the number of 1 bit that results from an operation?
 A) Zero B) Parity
 C) Auxiliary D) Carry
5. The addressing mode/s, which uses the PC instead of a general purpose register is _____
 A) Indexed with offset B) Relative
 C) Direct D) Both Indexed with offset and direct
6. What do you call the intermediate terms in binary multiplication?
 A) Multipliers B) Mid terms
 C) Partial Products D) Multiplicands
7. The number of sign bits in a 32-bit IEEE format is _____
 A) 1 B) 11
 C) 9 D) 23
8. The bits 1 & 1 are recorded as _____ in Booth's recoding.
 A) -1 B) 0
 C) +1 D) both -1 and 0
9. The classification of BUSES into synchronous and asynchronous is based on
 A) The devices connected to them B) The type of data transfer
 C) The Timing of data transfers D) None of the mentioned
10. The side of the interface circuits, that has the data path and the control signals to transfer data between interface and device is _____
 A) BUS side B) Port side
 C) Hardwell side D) Software side
11. _____ are used to overcome the difference in data transfer speeds of various devices.
 A) Speed enhancing circuitry B) Bridge circuits
 C) Multiple Buses D) Buffer registers

12. In _____ protocol the information is directly written into the main memory.
 A) Write through B) Write back
 C) Write first D) None of the mentioned
13. When dealing with multiple device interrupt, which mechanism is easy to implement
 A) Polling B) Interrupt nesting
 C) Vectored interrupt D) Daisy chain
14. The type of control signal is generated based on _____
 A) contents of the step counter B) Contents of IR
 C) Contents of condition flags D) All of the mentioned
15. What does the end instruction do?
 A) It ends the generation of a signal B) It ends the complete generation process
 C) It starts a new instruction fetch cycle and resets the counter D) It is used to shift the control to the processor
16. Which memory device is generally made of semiconductor
 A) RAM B) Hard disk
 C) Floppy disk D) CD drive
17. The interrupt-request line is a part of the _____
 A) Data line B) Control line
 C) Address line D) None of the mentioned
18. The signal sent to the device from the processor to the device after receiving an interrupt is _____
 A) Interrupt-acknowledge B) Return signal
 C) Service signal D) Permission signal
19. In DMA transfers, the required signals and addresses are given by the _____
 A) Processor B) Device drivers
 C) DMA controllers D) The program itself
20. A source program is usually in _____
 A) Assembly language B) Machine language
 C) High level language D) Natural language

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) How to measure the performance of a computer? Explain. Mention the basic performance equation with notation.	5	L*2	1	1
	b) Explain the steps involved in sequential multiplier algorithm and draw the sequential circuit multiplier diagram.	6	L2	1	1
	c) Explain single bus structure with neat diagram.	5	L2	1	1
2.	a) Write the steps involved in execution of instruction.	6	L2	1	1
	b) Define addressing modes. Explain the following addressing with syntax and example. i) Indexed addressing mode ii) Relative addressing mode	6	L2	1	1
	c) Perform multiplication -15×7 using bit pair recording multiplier. (Note: 7 is multiplier)	4	L3	2	1,2
3.	a) Explain the techniques to improve the processor performance.	6	L2	1	1
	b) Perform the division of $21/5$ using restoring division.	6	L2	2	1,2
	c) Explain IEEE notation for floating point numbers. Define normalization, overflow, and underflow.	4	L2	2	1
Unit – II					
4.	a) Describe Internal Organization of Semiconductor Memory Chips. Draw the internal organization of 16×8 memory chip.	8	L2	4	1,2

b)	Describe the execution of complete instruction using single bus organization with suitable diagram.	8	L2	3	1
5. a)	Compare hardwired control and microprogrammed control.	8	L3	3	1
b)	Explain the internal structure of synchronous DRAM with suitable diagram.	4	L2	4	1,2
c)	Write the control signals for the instruction Add (R3), R1. Considering single bus organization.	4	L2	3	1
6. a)	Explain the multi bus organization with neat diagram.	6	L2	3	1
b)	Describe the working of static RAM memories.	5	L2	4	1
c)	Define memory mapping. Explain associative and set associative mapping with neat diagram.	5	L2	4	1

Unit – III

7. a)	With neat diagrams explain the various methods to handle the multiple interrupt requests raised by multiple devices.	8	L2	5	1
b)	Define exception. Describe the different kinds of exceptions.	4	L2	5	1
c)	Explain the registers in Direct memory access (DMA) interface with neat diagram.	4	L2	5	1
8. a)	Define bus arbitration and describe the two approaches for bus arbitration with neat diagram.	8	L2	5	1
b)	Explain the synchronous and asynchronous bus with neat timing diagram.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
